MAGNET: Towards Adaptive GUI Agents with Memory-Driven Knowledge Evolution

Libo Sun, Jiwen Zhang, Siyuan Wang, Zhongyu Wei

Paper version: arXiv 2601.19199v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Frequent updates altering UI appearance and reorganizing workflows cause agents trained on historical data to fail.*

Existing memory systems store **text workflows**; MAGNET adds **visual** UI exemplars.

Motivation: two kinds of app drift

1. **Appearance drift.** Icons are redesigned.
2. **Workflow drift.** Steps are reorganized.
3. **Frozen knowledge.** Text-only memory.

Evaluation setup

AndroidWorld Easy · 61 | 3 offline datasets | Qwen2.5-VL-32B | Gemini-2.5-Pro | same action budget

- **Online** (AndroidWorld Easy) apart from **offline** step accuracy.
- **Not held out online:** memory built on the same 61 tasks.

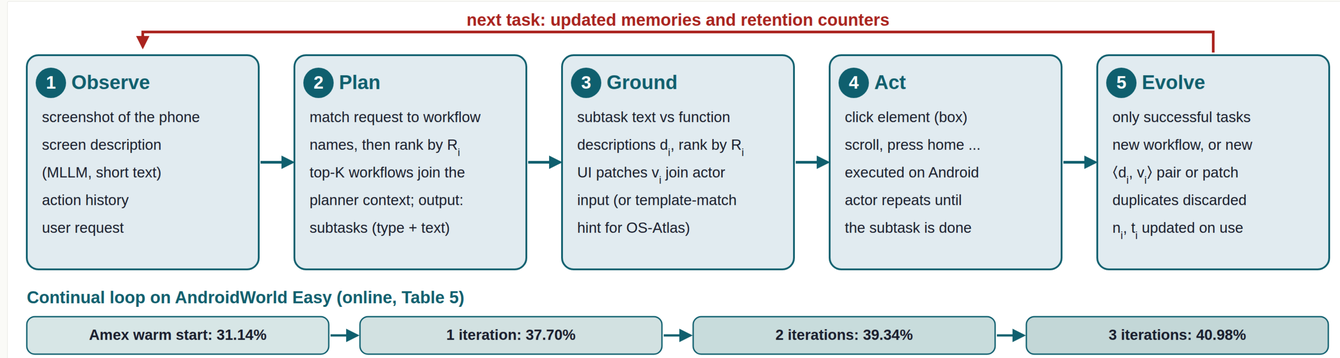
Our read (CUA Radar)

66	★★	online
score of 100	band	evidence level

- **Reports:** online eval; offline shift splits; same-backbone, same-budget baselines. **Not reported:** seeds, cost, code.
- **Flags:** AP5, memory data overlapping the evaluation.
- Pattern K27. Authors' numbers, not rerun.

Method: a planner–actor agent with two evolving memories

- **Sees / acts:** screenshot, screen text, history in; click (box), scroll, press home out.
- **Memory:** procedural workflows for the planner; stationary ⟨function, UI patch⟩ for the actor.
- **Signal:** successful runs only; entries ranked by usage and recency.



Our redraw (not the paper's figure): the planner–actor loop, the two memories and the evolution step.

$$R_i = \exp\left(-g_i / n_i\right), \quad g_i = C_{\text{global}} - t_i$$

retention score: gap since last access over retrieval count

Key result: AndroidWorld Easy online success (%)

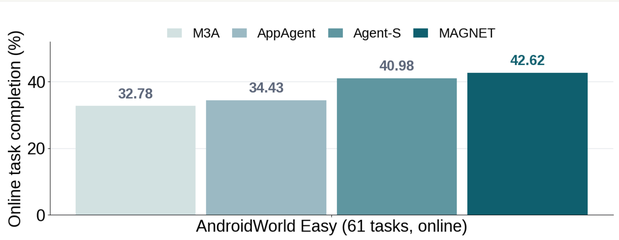


Table 2, redrawn: task completion, all Qwen2.5-VL-32B; no variance reported.

42.62% vs 40.98% Agent-S, same backbone

SO WHAT → *Offline step accuracy is not charted here.*

Ablation (offline): Amex in-distribution step SR (%)

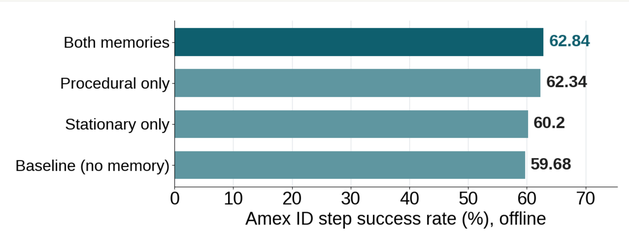


Table 3(a), redrawn: Qwen2.5-VL-32B as planner and actor; no online ablation.

SO WHAT → *Procedural memory gives +2.66 SR, stationary +0.52; online effect not ablated. Across five planner–actor pairs, memory adds +2.3 SR and +2.7 grounding on average (offline, three datasets). On shifted offline subsets, stationary memory gives +0.1 to +0.9 SR with Qwen; procedural memory ranges from slight degradation to gains.*

Headline Numbers

42.62%
AndroidWorld online SR
vs 40.98% Agent-S, same backbone

+1.6 pts vs Agent-S

+8.2 pts vs AppAgent

62.84% Amex step SR, offline

The paper's own controls

CONTROL	MAGNET	COMPARED WITH
Online memory source	3 passes, same 61 tasks	Agent-S: one round
Warm start to iter. 3	31.14 → 40.98 (online)	Amex share 26% / 18%
Offline AITZ step SR	43.50 / 52.77 (Gemini)	Atlas-Pro-7B 66.64 (tuned)

Takeaway

Dual memory reaches 42.62% online (Agent-S 40.98%) with small offline step gains; memory is built on the same 61 tasks, with no variance reported.

UI-40K: 41,009 entries | 20,618 unique intents | needs successful runs

Scan to read the full paper

arxiv.org/abs/2601.19199v2